

1. Administrative & Study Identification

Full Study Title: A Phase 3 Randomized, Open-label Study of MK-5684 Versus Alternative Abiraterone Acetate or Enzalutamide in Participants With Metastatic Castration-resistant Prostate Cancer (mCRPC) Previously Treated With Next-generation Hormonal Agent (NHA) and Taxane-based Chemotherapy (OMAHA-003) **Short Title/ Acronym:** OMAHA-003 / MK-5684-003 **Protocol Number:** MK-5684-003 **NCT Number:** NCT06136624 **Trial Status:** RECRUITING **Sponsor Name:** Merck Sharp & Dohme LLC (Merck) **Investigational Product:** Opevesostat (MK-5684) vs. Alternative NHA (Abiraterone acetate or Enzalutamide) **Phase of Development:** Phase 3 **Medical Monitor:** Merck Clinical Operations Lead **Disease Area:** Metastatic Prostate Cancer (mCRPC)

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of opevesostat (MK-5684) compared to an alternative NHA (abiraterone acetate or enzalutamide) with respect to Overall Survival (OS) in androgen receptor ligand binding domain (AR-LBD) mutation-positive and AR-LBD mutation-negative participants with metastatic castration-resistant prostate cancer (mCRPC).

Secondary Objectives: To evaluate radiographic Progression-Free Survival (rPFS) and Time to Initiation of the First Subsequent Anti-Cancer Therapy or Death (TFST). Additional objectives include evaluating Objective Response Rate (ORR), Duration of Response (DOR), safety, and tolerability.

2.2 Background & Rationale Androgen receptor (AR) signaling remains a key driver of prostate cancer even after progression on androgen-directed therapies. Activating AR ligand binding domain (AR-LBD) somatic point mutations are a common mechanism of resistance to standard treatments. Opevesostat (MK-5684) is an oral, nonsteroidal inhibitor of cytochrome P450 11A1 (CYP11A1), which catalyzes the first and rate-limiting step of steroid biosynthesis. Its inhibition suppresses the production of all steroid hormones, providing a novel approach to overcome resistance in patients with heavily pretreated mCRPC.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Alternative NHA) **Blinding:** Open-label **Randomization:** 1:1 Randomized

3.2 Planned Enrollment & Duration Target \$\$\$: Approximately 1,310 participants **Number of Sites:** Global locations **Estimated Study Start:** 31-Dec-2023 **Estimated Primary Completion:** 02-Aug-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male participants with histologically- or cytologically-confirmed adenocarcinoma of the prostate without small cell histology. 2. Prostate cancer progression while on androgen deprivation therapy (ADT) or post bilateral orchiectomy within 6 months before Screening. 3. Current evidence of distant metastatic disease (M1 disease) documented by bone lesions on bone scan and/or soft tissue disease by CT/MRI. 4. Disease progressed during or after treatment with 1 novel hormonal agent (NHA). 5. Received 1 but no more than 2 taxane-based chemotherapy regimens for mCRPC with progressive disease (PD) during or after treatment. 6. Ongoing ADT with serum testosterone <50 ng/dL (<1.7 nM). 7. Must provide tumor tissue from a fresh core or excisional biopsy from soft tissue not previously irradiated. 8. ECOG performance status of 0 or 1 assessed within 7 days of randomization. 9. Prior treatment with PARPi (or deemed ineligible/refused). 10. Prior 177Lu-PSMA-617 (or deemed ineligible/refused). 11. Participants who have not received cabazitaxel can enroll if ineligible or refused treatment. 12. If treated with first-generation anti-androgens before screening, evidence of PD >4 weeks since last flutamide and >6 weeks since last bicalutamide/nilutamide. 13. Bone resorptive therapy (e.g., bisphosphonate/denosumab) must be on stable doses for ≥ 4 weeks before randomization. 14. HIV infection is permitted if well-controlled on antiretroviral therapy (ART). 15. HBV is permitted if on antiviral therapy for ≥ 4 weeks with undetectable viral load. HCV permitted if viral load is undetectable. 16. Strict contraceptive adherence required (condom or abstinence) during and post-treatment (7 days for opevesostat, 30 days for abiraterone, 3 months for enzalutamide).

4.2 Exclusion Criteria 1. Gastrointestinal disorder that might affect absorption, or unable to swallow capsules/tablets. 2. History of pituitary dysfunction or poorly controlled diabetes mellitus. 3. Clinically significant abnormal serum potassium or sodium levels. 4. History of active/unstable cardio/cerebro-vascular disease (including thromboembolic events), clinically significant ventricular arrhythmias, long QTc syndrome, hypotension (<110 mm Hg), or uncontrolled hypertension ($\geq 160/90$ mm Hg). 5. History of seizure within 6 months or predispositions to seizures within 12 months. 6. "Superscan" bone scan. 7. Active CNS metastases and/or carcinomatous meningitis. 8. Known additional malignancy progressing or requiring active treatment in the past 3 years. 9. Active autoimmune disease requiring systemic treatment in past 2 years, active infection requiring systemic therapy, or concurrent active HBV/HCV. 10. Diagnosis of immunodeficiency, or receiving chronic systemic

steroids/immunosuppressives within 7 days prior to first dose. 11. Major surgery (except prostate biopsy) within 28 days before randomization and not adequately recovered. 12. Recent therapies prior to randomization: Anticancer mAb (within 4 weeks), herbal/medicinal products decreasing PSA (within 4 weeks), Radium-223/Lutetium-177 (within 4 weeks), 5- α reductase inhibitors/estrogens/cyproterone (within 4 weeks), colony-stimulating factors (within 28 days), whole blood transfusion (within 120 days), prior targeted small molecule/NHA (enza/apa within 3 weeks; abi/daro within 2 weeks), systemic anticancer therapy (within 4 weeks), radiotherapy (within 2 weeks). 13. Received a live or live-attenuated vaccine within 30 days before the first dose. 14. Unstable dose of thyroid hormone therapy within 6 months. 15. Systemic use of strong CYP3A4 inducers, P-gp inhibitors, or CYP450-inducing antiepileptic drugs within 2 weeks before the first dose. 16. Use of aldosterone antagonists (e.g., spironolactone) and phenytoin within 4 weeks.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Arm 1: Opevesostat 5 mg PO BID (+ dexamethasone 1.5 mg QD + fludrocortisone 0.1 mg QD). Arm 2: Enzalutamide 160 mg PO QD OR Abiraterone acetate 1000 mg PO QD (+ prednisone/corticosteroid management). **Route of Administration:** Oral **Duration of Treatment:** Continues until radiographic disease progression, unacceptable toxicity, investigator's decision, or withdrawal of consent.

5.2 Concomitant & Prohibited Therapy Permitted: LHRH agonist/antagonist for ongoing androgen deprivation, bone resorptive therapies (if on stable doses for $\geq$ 4 weeks prior). **Prohibited:** Strong CYP3A4 inducers, P-gp inhibitors, herbal products known to decrease PSA, radium-223, lutetium-177, and aldosterone antagonists.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Days -28 to 0	Informed Consent, Medical History, Imaging, AR-LBD Mutation Testing, Viral/Lab Screening
Baseline	Day 0	Randomization, First Dose, EKG, Vital Signs
Interim	Every 2 weeks (first 3 mos), then Q4W	Safety Labs, PSA Evaluation, Tumor Imaging, AE Monitoring
End of Study	Upon progression	Exit Interview, IP Accountability, Follow-up Phase for Survival

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. **Overall Survival (OS) in AR-LBD Mutation-Positive Participants:** Defined as the time from

randomization to death due to any cause. 2. **Overall Survival (OS) in AR-LBD Mutation-Negative Participants:** Defined as the time from randomization to death due to any cause.

7.2 Secondary & Exploratory Endpoints

- 1. Radiographic Progression-Free Survival (rPFS) in AR-LBD Mutation-Positive Participants:** Assessed by Blinded Independent Central Review (BICR) per Prostate Cancer Working Group-modified RECIST 1.1.
- 2. Radiographic Progression-Free Survival (rPFS) in AR-LBD Mutation-Negative Participants:** Assessed by BICR per PCWG-modified RECIST 1.1.
- 3. Time to Initiation of the First Subsequent Anti-Cancer Therapy or Death (TFST).** * Exploratory: Objective Response Rate (ORR), Duration of Response (DOR), time to first symptomatic skeletal-related event, PSA progression, and pain progression.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Assessed continuously and graded according to NCI CTCAE v5.0 criteria via patient reporting and clinical labs. **Serious Adverse Events (SAEs):** Reported to the sponsor within standard 24-hour reporting timelines. **Data Monitoring Committee (DMC):** External independent safety committee will monitor the trial parameters continuously.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intention-to-Treat (ITT) population for primary efficacy assessments; Safety Set for AE/SAE metrics. **Sample Size Justification:** \$N \approx 1,310\$ powered to detect statistically significant improvements in OS and rPFS across both AR-LBD mutation-positive and -negative biomarker subgroups. **Handling of Missing Data:** Standard censoring rules for time-to-event endpoints; OS censored at the date the participant was last known to be alive.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Comprehensive onsite visits combined with remote/central data monitoring per the sponsor's clinical monitoring protocol. **Audit Trail:** Secure eCRF locking processes and standard electronic data clarification forms (DCFs).

11. Study Identifiers & Governance

Primary ID: MK-5684-003 **Registry Number:** NCT06136624 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC **Study Title:** Study of

Opevesostat (MK-5684) Versus Alternative NHA in mCRPC (MK-5684-003) **Official Regulatory Title:** A Phase 3 Randomized, Open-label Study of MK-5684 Versus Alternative Abiraterone Acetate or Enzalutamide in Participants With Metastatic Castration-resistant Prostate Cancer (mCRPC) Previously Treated With Next-generation Hormonal Agent (NHA) and Taxane-based Chemotherapy (OMAHA-003)

12. Clinical Framework (mCRPC Specific)

Primary Condition: Metastatic Castration-resistant Prostate Cancer (mCRPC) **Diagnosis Threshold:** Distant metastatic disease (M1) progressing on ADT and 1 prior NHA + 1-2 prior taxanes **Medical Methodology:** CYP11A1 inhibition vs alternative NHA (abiraterone/enzalutamide) **Optimization Target:** Target inhibition of all steroid hormone precursors driving AR signaling

13. Study Procedures & Methodology

Management Pathway: Oral targeted therapy clinical pathway for mCRPC **Education Protocol (HCP):** Site initiation visits and investigator training regarding CYP11A1 mechanism and AE tracking **Education Protocol (Patient):** Onsite training for oral drug administration, diary documentation, and concomitant drug restrictions **Quality Audit Mechanism:** Regular central data monitoring and rigorous clinical site auditing

14. Observation & Follow-Up Schedule

Total Duration: Estimated up to 4.5 years (approx. 54 months) **Onsite Visit Cadence:** Every 2 weeks for the first 3 months, then every 4 weeks **Remote Monitoring Frequency:** Between onsite visits as clinically indicated **Checklist Review Interval:** Every 4 weeks (Q4W) during active treatment

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				